1. 5A's — Steps in the EBM Process

The five steps are:

1. **Assess the patient.** A clinical problem or question arises from the care of the patient.
2. **Ask the question.** Construct a well-built clinical question.
3. **Acquire the evidence.** Select the appropriate resources and conduct a search.
4. **Appraise the evidence.** Check validity and applicability.
5. **Apply the evidence.** Integrate the evidence with clinical expertise and patient preferences, and apply it to practice.

Two further steps follow these five. **Analyze outcomes:** determine whether the evidence-based action was effective in practice. **Add knowledge:** convert the clinical experience and the outcome into knowledge for later decisions.

2. PICOS Model

PICOS breaks a clinical question into the parts needed for a search.

- **P — Patient, population, or problem:** the features that make the group like the patient being treated.
- **I — Intervention, exposure, or prognostic factor:** the drug, test, exposure, or factor under study.
- **C — Comparison:** the main alternative, usually usual care. When the question names no comparison, C is "not stated".
- **O — Outcome:** the specific benefit or harm that is measured.
- **S — Study design:** the design that can answer the question. A treatment question calls for a randomized trial. A prognosis question calls for a cohort study.

PICO states the question and supplies the search terms. S states which studies are eligible. Quantitative synthesis uses PICO. Qualitative and mixed-methods synthesis uses SPIDER: Sample, Phenomenon of Interest, Design, Evaluation, and Research type.

3. Categories of Evidence

Evidence falls into four categories:

1. **Studies:** unfiltered original studies, such as those in MEDLINE and PubMed.
2. **Summaries:** systematic reviews, such as Cochrane reviews.
3. **Synopses:** pre-appraised journals, such as ACP Journal Club.
4. **Systems:** evidence built into a decision resource, such as Clinical Evidence and UpToDate.

The same evidence is also arranged in the 5S hierarchy. From the top:

- **Systems:** computerized decision support.
- **Summaries:** evidence-based textbooks, such as UpToDate, DynaMed, ACP PIER, and BMJ Clinical Evidence.
- **Synopses:** evidence-based journals, such as *EBM*, *EBN*, *EBMH*, and ACP Journal Club.
- **Syntheses:** systematic reviews, such as Cochrane reviews.
- **Studies:** original articles, such as those found through PubMed and Clinical Queries.

A search starts at Systems and moves down the hierarchy. In the four categories, Summaries are systematic reviews. In the 5S hierarchy, Summaries are evidence-based textbooks, and systematic reviews are Syntheses.

4. Benefits of Adopting EBM

Adopting evidence-based medicine has three benefits:

1. It reduces errors and improves the quality of patient care.
2. It reduces the cost of treatment.
3. It advances knowledge and keeps practice up to date with scientific progress.

Evidence-based medicine also makes it possible to change practice and to improve treatment. A decision combines the current best evidence, clinical expertise, and the patient's values, and applies them to the individual patient.

5. Key Steps for Systematic Review and Meta-Analysis

The main steps are:

1. Define the review question.
2. Search for studies, using a documented search strategy.
3. Screen the studies.
4. Select the studies.
5. Extract the data.
6. Analyse the data. This step is the meta-analysis when pooling is appropriate.
7. Report the review.
8. Update the review.

These eight steps are carried out as follows:

1. **Research question.** The question should be feasible, interesting, novel, ethical, and relevant. Quantitative reviews use PICO. Qualitative and mixed-methods reviews use SPIDER.
2. **Preliminary search.** Check whether the review has already been done, and estimate how many studies are likely to be included.
3. **Inclusion and exclusion criteria.** Set the criteria before searching.
4. **Search strategy.** The strategy states the databases, search terms, Boolean operators, and limits, so that another person can repeat the search.
5. **Database search.** The databases include PubMed, EMBASE, Google Scholar, Scopus, the Cochrane Library, and other relevant sources. AMSTAR requires at least two databases. A systematic review searches at least three. The strategy uses AND, OR, and NOT.
6. **Protocol and registration.** The protocol is registered before the search begins. Registration makes the process transparent and prevents duplication. It records the question, eligibility criteria, intervention or exposure, quality assessment, and the planned analysis. After revision by the principal investigator, it is registered with PROSPERO or a Cochrane or Campbell registry.
7. **Title and abstract screening.** Two or three reviewers include or exclude each record according to the criteria.
8. **Full-text screening.** Download the full texts, apply all criteria, and record the reason for each exclusion.
9. **Manual search.** Check the references of included studies and existing reviews, contact authors and experts, and look at related and cited articles.
10. **Data extraction and quality assessment.**
11. **Statistical analysis.** Pool the studies when they are similar enough. This is the meta-analysis.

12. **Writing, revision, and submission.** Report the review with the PRISMA checklist and flow diagram.

A systematic review remains valid when the studies cannot be pooled. The results are then presented in tables or text. A meta-analysis needs at least two studies that are similar enough. The pooled estimate is shown on a forest plot. The team should include at least three people: a content expert, a review-methods expert, and a statistician. The protocol is registered before the review starts.

6. Types and Functions of Boolean Logic Terms

Boolean operators set the relationship between search concepts and change which records are retrieved.

- **AND** retrieves only articles that contain all of the terms. It narrows the search. Example: “child obesity” AND Maryland.
- **OR** retrieves articles that contain any of the terms. It broadens the search. Example: “child obesity” OR “overweight children”.
- **NOT** removes articles that contain the second term, even when the first term is also present. It narrows the search. Example: obesity AND children NOT adults.

Synonyms within one concept are combined with OR. Different concepts are combined with AND. Parentheses set the order when more than one operator is used, for example `(children AND teenagers) AND (ADHD OR hyperactivity)`. Quotation marks keep a phrase together, for example “eating disorder”.

7. The GRADE Assessment System’s 3 Upward Factors and 5 Downward Factors

GRADE expresses how much confidence can be placed in an estimate of effect. A body of randomized trials starts at high certainty. All other study designs start at low certainty. Randomized and observational evidence are rated separately.

The four levels are high (++++), moderate (+++), low (++), and very low (+).

Five factors can lower certainty, by one level when the problem is serious and by two levels when it is very serious:

1. **Risk of bias:** limitations in design or conduct that affect the outcome of interest.
2. **Inconsistency:** differences in the direction or size of the effect, or a large I^2, that are not explained.
3. **Indirectness:** populations, interventions, comparisons, or outcomes that differ from the question, including surrogate outcomes and the absence of a direct comparison.
4. **Imprecision:** few participants, few events, or a confidence interval too wide to be informative, such as an interval that includes a risk ratio of 1.
5. **Publication bias:** selective publication that pushes the estimate upward or downward. Clues include small studies, industry sponsorship, and an asymmetric funnel plot.

Three factors can raise certainty:

1. **Large effect.** A relative risk reduction of 0.5 or less, or a risk ratio of 2 or more, raises certainty by one level. A relative risk reduction of 0.8 or less, or a risk ratio of 5 or more, raises it by two levels.
2. **Dose–response gradient,** which raises certainty by one level.
3. **Favourable direction of residual confounding,** which raises certainty by one level. All plausible residual confounding would have reduced an effect that was observed, or would have produced an effect when none was observed.

Certainty of evidence and strength of a recommendation are separate judgements. High-certainty evidence can support a weak recommendation, and very low-certainty evidence can support a strong recommendation.

8. Possible Methods and Graphs Used in Quantitative Analysis of Meta-Analysis

A meta-analysis is the weighted average effect of at least two studies that are similar enough to combine. The result is shown on a forest plot. Studies that should not be combined are reported in a table or in text.

Effect measures. For dichotomous outcomes, the measures are the odds ratio, risk ratio, and risk difference. For continuous outcomes, they are the mean difference and the standardised mean difference. The size and direction of an effect depend on the sample size, the variance, and the reliability of the outcome measure.

Models.

- A **fixed-effect** model assumes one true effect. Observed differences are random error. It is used when the studies have the same design and methods and within-study variation is small. Studies are weighted by within-study variance only, so larger studies receive more weight and the confidence interval is narrower.
- A **random-effects** model assumes that the true effect can differ from study to study. It is used when the studies are expected to differ, even if the heterogeneity test is not significant, and when the result is meant to apply across settings. The weight includes both within-study and between-study variance, so the confidence interval is wider.

Methods of weighting.

- Fixed effect: inverse variance when there are few studies and each is large; the Mantel–Haenszel method when there are many small studies; the Peto method for an odds ratio when the event rate is low or a cell is zero.
- Random effects: the DerSimonian–Laird method, mainly for dichotomous data, and inverse-variance weighting for continuous data.

Further analyses when heterogeneity needs explanation. Sensitivity analysis repeats the analysis after an assumption is changed. Subgroup analysis splits the studies on a characteristic chosen in advance. Meta-regression uses the effect size as the outcome and study-level characteristics as the predictors. Each study is one observation.

Graphs.

- A **forest plot** shows each study's effect, confidence interval, and weight, together with the pooled effect. The line of no effect is 0 for a difference and 1 for a ratio.
- A **funnel plot** plots the effect against precision and is used to look for publication bias.
- An **L'Abbé plot** plots the control event rate against the experimental event rate. A wide scatter suggests heterogeneity.
- A **Galbraith plot** plots precision against the standardised effect. Scatter around the regression line shows outlying studies.
- A **Baujat plot** plots each study's contribution to Cochran's Q against its influence on the pooled effect.
- A **bubble plot** shows a meta-regression of the effect size on a study-level covariate.

9. Cochrane's Position on the Use of AI

Cochrane accepts the use of artificial intelligence, provided it does not reduce the integrity, quality, or reliability of the review. Human authors remain accountable.

Since 2016, Cochrane has used machine learning to identify randomized trials. A classifier is used in Cochrane reviews only when its recall is about 99%, because missing a relevant study is not acceptable.

Authors are responsible for the content, methods, and findings. This includes the decision to use an AI tool, the way it is used, and the effect it has on the review. Use of a tool must be justified, and the tool must have been evaluated for that type of review.

An AI tool is not an author and cannot meet the criteria for authorship. It must not be used to fabricate or falsify data, results, or findings. A chatbot is not a knowledge base and should not be relied on for facts.

Whenever AI makes or suggests a judgement, that use is reported. This includes decisions about eligibility, risk of bias, data extraction, synthesis, and GRADE. The report gives the name, version, and date of the tool, its purpose, its validation, the human check of its output, any interests and funding, and its limitations. Automation tools are also reported under PRISMA.

10. Heterogeneity: Types and Detection Methods

Heterogeneity is variability among the studies in a systematic review. Assessing it means judging how far the studies differ and whether that difference can be measured, and then deciding whether and how to combine them. Some difference is always present. Only a difference large enough to change the conclusion calls for a change in how the studies are combined.

Types.

- **Clinical heterogeneity** is a difference that affects whether studies can be compared or combined. It includes the participants' characteristics, the severity of disease, the setting, the frequency and intensity of the intervention, the biological characteristics of the participants, the type of intervention, and the outcomes measured.
- **Methodological heterogeneity** is a difference in risk of bias, study design, or measurement tools.
- **Statistical heterogeneity** is a difference in effect size among studies that is greater than chance.

Detection.

On a forest plot, point estimates that lie far apart and confidence intervals that do not overlap suggest heterogeneity. Greater overlap means less heterogeneity. Heterogeneity can also be seen on an L'Abbé plot, a Galbraith plot, a Baujat plot, and a funnel plot.

- **Cochran's Q** is the traditional chi-square test. A large Q means wider variation around the average effect. The degrees of freedom equal the number of studies minus 1. A p value below 0.05 indicates heterogeneity. A p value above 0.05 indicates homogeneity on this test. The test has low power when there are few studies.
- **I^2** is the percentage of the observed variation that comes from real differences rather than chance: $I^2 = 100\% \times (Q - df)/Q$. An I^2 of 20% means that 20% of the observed variation in treatment effects cannot be attributed to chance alone. About 25% is low, 50% is moderate, and 75% or more is high. I^2 does not identify the cause.
- **τ^2** is the variance of the true effect sizes. It is used to weight the studies in a random-effects model. A value near zero means little between-study variance. τ, the standard deviation of that distribution, is used to calculate the prediction interval.

When heterogeneity is high, the results may be left unpooled. A fixed-effect model is appropriate only when the studies share the same design and methods and the variation is random error. Otherwise a random-effects model includes the between-study variance. The source of heterogeneity is examined by meta-regression, a subgroup analysis specified in advance, or a sensitivity analysis. In meta-regression, each study is one observation.